Journal of the Optical Society of America B
OPTICAL PHYSICS

Efficient sideband generation and beam shaping in strongly coupled graphene-based spatiotemporal metasurfaces

O. Poordashtban, M. M. Sadafi, and H. Mosallaei*

Northeastern University, Department of Electrical Engineering, 360 Huntington Ave, Boston, Massachusetts 02115, USA
**h.mosallaei@northeastern.edu*

Received 16 April 2026; revised 30 May 2026; accepted 6 June 2026; posted 8 June 2026; published 7 July 2026

Achieving continuous $0-2\pi$ phase modulation with uniform reflection amplitude in a single-gate electrically tunable mid-infrared metasurface remains a fundamental challenge. Here, we demonstrate that strong coupling between an overcoupled guided-mode resonance (GMR) and an electrically tunable graphene plasmonic (GP) mode overcomes this limitation. Electrical tuning of the graphene Fermi level drives strong mode hybridization, enabling continuous $\sim 2\pi$ phase modulation with a minimum reflection amplitude of 0.57 at $\lambda = 6.83$ μm. An inverse-design optimization framework compensates for inter-cell coupling, improving beam-steering efficiency from 57% to 85%. We further demonstrate spatiotemporal modulation enabling single-sideband suppressed-carrier beam steering with a harmonic conversion efficiency of ∼87% and a first-order diffraction efficiency of ∼97%. These results establish a scalable platform for programmable mid-infrared wavefront engineering with applications in beam steering, holography, and optical communications.

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https://doi.org/10.1364/JOSAB.601458

1. INTRODUCTION

Precise and electrically reconfigurable control of optical wavefronts at the single-pixel level is a foundational capability for many emerging photonic technologies. This capability is particularly important in the mid-infrared (mid-IR) spectral range, which spans the molecular fingerprint region (2.5–25 μm) [1], atmospheric transmission windows [2], and free-space communication bands [3]. Dynamic mid-IR wavefront engineering underpins applications including beam steering and LiDAR for autonomous systems [4–7], reconfigurable holographic displays and augmented or mixed-reality imaging [8,9], free-space optical communications [3], active computational metasurface architectures for optical analog processing and machine-learning hardware [10–13], and label-free molecular fingerprinting and spectroscopy [14,15]. Beyond static wavefront shaping, metasurfaces capable of spatiotemporal modulation, where both the spatial phase profile and the temporal modulation frequency are dynamically programmed, enable functionalities such as single-sideband suppressed-carrier (SSB-SC) modulation [16], nonreciprocal wavefront manipulation [17], and frequency-converted beam steering [16,17]. Realizing these capabilities within a single electrically scalable mid-IR platform remains an outstanding challenge.

Full wavefront synthesis requires continuous and reversible phase modulation spanning the complete $0-2\pi$ range while maintaining high and spatially uniform scattering amplitude. Achieving this condition in electrically tunable metasurfaces remains fundamentally challenging. In resonant meta-atoms, tuning the resonance typically modifies both the phase and the amplitude of the scattered field simultaneously, producing intrinsic phase–amplitude coupling. As a result, the complex reflection coefficient follows a constrained trajectory in the complex plane; tuning typically moves the response along a limited arc rather than completing a full circle, preventing full 2π phase coverage while maintaining a constant amplitude. This difficulty is further compounded by optical losses in active materials near resonance, where absorption suppresses reflectance precisely in the phase-sensitive spectral region. In addition, the refractive-index modulation achievable in most gate-tunable materials is relatively small, restricting the achievable resonance shift and therefore limiting the accessible phase range to well below 2π [18,19]. Consequently, achieving complete $0-2\pi$ phase modulation with uniform amplitude remains difficult in single-gate tunable metasurface platforms.

One promising strategy to overcome this limitation is to exploit hybridization between two resonant modes with complementary properties. When two resonances are strongly coupled, their interaction produces an avoided-crossing behavior in which the phase response can be redistributed through

mode hybridization while preserving amplitude. Several strategies have explored this concept. Multi-gate metasurfaces, in which independent electrical channels control separate degrees of freedom, have demonstrated that 2π phase modulation is achievable [16,20]. However, the introduction of multiple control electrodes per pixel increases fabrication complexity and places stringent requirements on driving circuitry, reducing scalability. In the mid-IR regime, graphene-integrated plasmonic metasurfaces have demonstrated electrically tunable phase modulation approaching 237° [21], but still fall short of the full 2π range required for arbitrary wavefront synthesis. A conceptually significant advance was recently reported by Kim *et al.* [22], who demonstrated that avoided crossing between a quasi-bound state in the continuum (qBIC) resonance and a graphene plasmonic (GP) mode enables full 2π phase modulation within a single-gate platform while maintaining nearly constant reflection amplitude.

Despite this success, qBIC-based metasurfaces exhibit a fundamental limitation for scalable wavefront engineering: the strong nonlocality associated with ultra-high-Q resonances. Both qBIC and guided-mode resonances (GMRs) arise from spatially extended, lattice-supported collective modes [23,24]. However, the spatial coherence length of these modes increases with the resonance quality factor. Ultra-high-Q qBIC resonances therefore exhibit long photon lifetimes and large spatial coherence lengths, significantly enhancing inter-pixel cross-talk in metasurface arrays [23]. This nonlocal interaction complicates independent pixel-level phase control and limits scalability. Moderate-Q guided-mode resonances provide a more favorable balance between strong resonant phase response and reduced spatial coherence, enabling improved pixel independence while still supporting strong phase modulation.

Graphene provides a uniquely suitable electrically active material for tunable metasurfaces operating in the mid-infrared. Unlike noble metals, whose plasmonic resonances are fixed by geometry, graphene supports electrically tunable surface plasmons intrinsically in the mid-IR and terahertz spectral regimes [25–27]. The resonance frequency of graphene plasmons scales approximately as $\omega_{\mathrm{GP}} \propto \sqrt{E_f}$ [28,29], enabling large and continuous spectral tuning through a single gate voltage. In addition, graphene plasmons exhibit extreme modal confinement, orders of magnitude tighter than the free-space wavelength, which produces strong field overlap with nearby dielectric resonances and facilitates efficient photonic–plasmonic mode coupling. Graphene also supports high-speed electro-optic modulation, with experimentally demonstrated bandwidths exceeding 30 GHz [30]. These properties make graphene an attractive platform for electrically tunable mid-IR metasurfaces.

Here, we demonstrate that avoided crossing between an overcoupled guided-mode resonance and a graphene plasmonic mode provides an efficient pathway to achieve electrically tunable 2π phase modulation while reducing the spatial nonlocality associated with ultra-high-Q resonances (see Note S7 of Supplement 1). Specifically, we engineer a metasurface operating under TM polarization at $\lambda = 6.83$ μm in which an overcoupled GMR interacts with a graphene plasmonic mode. The GMR is designed in the overcoupled regime, where the radiative decay rate γ_r exceeds the intrinsic dissipative loss rate γ_d, ensuring that mode hybridization redistributes the phase rather than suppressing the amplitude. Electrical tuning of the graphene Fermi level shifts the GP resonance across the fixed GMR frequency, inducing strong hybridization and phase repulsion. This mechanism enables continuous $\sim 2\pi$ phase modulation with a minimum reflection amplitude of 0.57 across the entire tuning range.

The moderate-Q nature of the GMR ($Q_{\mathrm{GMR}} \approx 480$) significantly reduces the spatial coherence length compared with qBIC resonances, providing a more favorable starting point for independent pixel addressing. Residual nonlocal coupling is further mitigated through an inverse-design optimization framework that tailors the geometric parameters of individual unit cells to compensate for inter-cell interactions. Using this approach, we achieve approximately 28% improvement in beam-steering efficiency compared with a conventional forward-designed metasurface. Building on the demonstrated 2π phase control and near-uniform reflectance, we further demonstrate single-sideband suppressed-carrier modulation with a frequency-conversion efficiency of $\eta_{\mathrm{SSB}} \approx 87\%$. These results establish a scalable pathway toward programmable mid-infrared metasurfaces capable of dynamic beam steering, holography, free-space communications, and optical computing.

The key contributions of this work are threefold. First, we demonstrate that avoided crossing between an overcoupled guided-mode resonance (GMR) and a graphene plasmonic (GP) mode enables continuous electrically tunable 2π phase modulation with relatively uniform reflection amplitude in the mid-infrared. Second, we show that the moderate-Q nature of the GMR enables pixel-level phase programmability in metasurface arrays. Then, using an inverse-design optimization framework, we compensate for residual inter-pixel coupling, which significantly improves beam-steering efficiency. Third, leveraging the demonstrated 2π phase tunability and uniform reflectance, we further realize spatiotemporal modulation enabling single-sideband suppressed-carrier beam steering.

2. RESULTS AND DISCUSSION

Our approach relies on controlled coupling between two resonant modes with complementary properties: an overcoupled, spectrally narrow resonance and a second resonance exhibiting large frequency tunability. When these resonances possess significant spatial field overlap, their near-field interaction produces a finite coupling strength, resulting in an avoided-crossing behavior in the spectral response. The resulting hybridized eigenmodes inherit the key characteristics of both parent resonances: the strong radiative coupling of the overcoupled resonance and the large spectral tunability of the second mode. This hybridization mechanism enables large phase excursions while limiting amplitude variation over the tuning range, which is a key requirement for high-fidelity wavefront control.

In the proposed metasurface, the spectrally narrow resonance is provided by a dielectric guided-mode resonance (GMR) supported by a silicon grating structure. Dielectric resonances exhibit intrinsically low dissipative loss, while their radiative loss rate can be engineered through geometric design.

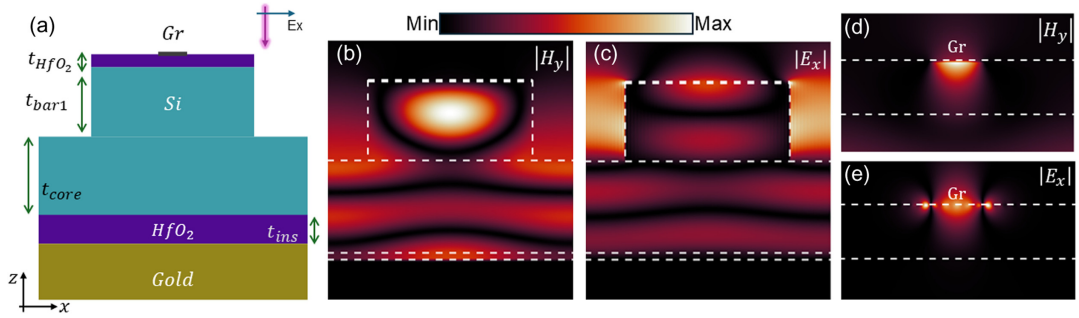


Fig. 1. Device architecture and resonant field distributions of the hybridized GMR–graphene metasurface. (a) Cross-sectional schematic of the metasurface unit cell consisting of a silicon grating supported by a multilayer dielectric stack and backed by a gold mirror. (b),(c) Simulated near-field distributions of the guided-mode resonance (GMR) of the silicon grating under TM illumination. Panel (b) shows the magnetic field magnitude $|H_y|$, while (c) shows the electric field magnitude $|E_x|$. In this case, the graphene plasmonic (GP) mode is off-resonant. (d),(e) Simulated near-field distributions of the graphene plasmonic (GP) resonance. Panel (d) shows $|H_y|$ and (e) shows $|E_x|$. Here the GMR is slightly detuned while the GP mode is resonant. The spatial overlap between the dielectric GMR field and the graphene plasmonic field indicates strong modal co-localization, enabling efficient coupling between the two resonances through electrical tuning of the graphene Fermi level E_f.

By tuning the grating parameters, the radiative decay rate ($\gamma_r = 3.7 \times 10^{10}$ Hz) can be made larger than the dissipative loss rate ($\gamma_d = 1.1 \times 10^{10}$ Hz), placing the resonance in the overcoupled regime (see Note S5 of Supplement 1). In this regime, the phase of the reflection coefficient varies rapidly, enabling large phase modulation.

The second resonance is provided by a graphene plasmonic (GP) mode. Graphene plasmons are well known for their strong electric field confinement, an extremely small modal volume resulting from the two-dimensional nature of graphene, and substantial electrical tunability through modulation of the graphene Fermi level. These properties make graphene plasmons an ideal candidate for the highly tunable resonance required for the hybridization mechanism.

The proposed metasurface structure is shown in Fig. 1(a). The device operates in reflection mode and consists of a periodic silicon bar array supported by a multilayer dielectric stack backed by a gold mirror. From bottom to top, the structure consists of a gold back reflector, an optical insulator layer of thickness $t_{\text{ins}} = 145$ nm, a silicon guiding core of thickness $t_{\text{core}} = 1954$ nm, a silicon bar of width $w_{\text{bar}} = 1766$ nm and height $t_{\text{bar}} = 1663$ nm, a thin gate dielectric layer (HfO_2) of thickness $t_{HfO2} = 30$ nm, and a patterned monolayer graphene strip of width $w_g = 38$ nm. The structure is periodic with period $\Lambda = 2596$ nm. The final geometry was obtained through numerical optimization using the rigorous coupled-wave analysis (RCWA) method to simultaneously maximize the achievable phase excursion while maintaining nearly uniform reflectance across the tuning range. Further details of the optimization procedure are provided in Supplement 1.

Under TM-polarized normal incidence, the silicon grating supports a guided-mode resonance with strong electric-field localization near the top interface of the silicon guiding core, as shown in Fig. 1(c). The graphene strip is intentionally positioned at this field maximum to maximize the modal overlap between the GMR and the graphene plasmonic mode. The simulated electromagnetic field distributions shown in Figs. 1(b)–1(e) confirm the strong spatial co-localization of the dielectric resonance and the graphene plasmonic mode, which is required for efficient mode coupling.

The coupled system can be described using a two-mode model derived from temporal coupled-mode theory (TCMT) [22,31], which can be expressed in the form of an effective non-Hermitian Hamiltonian:

$$H = \begin{pmatrix} \omega_{\text{GMR}} & u \\ u & \omega_{\text{GP}}(E_f) \end{pmatrix} + i \begin{pmatrix} \gamma_{\text{GMR},d} + \gamma_{\text{GMR},r} & \sqrt{\gamma_{\text{GMR},r}\gamma_{\text{GP},r}} \\ \sqrt{\gamma_{\text{GMR},r}\gamma_{\text{GP},r}} & \gamma_{\text{GP},d} + \gamma_{\text{GP},r} \end{pmatrix}, \quad (1)$$

where ω_{GMR} and $\omega_{\text{GP}}(E_f)$ denote the uncoupled resonance frequencies of the GMR and GP modes, respectively, and u represents the near-field coupling strength between the two resonances. The parameters γ_d and γ_r correspond to the dissipative and radiative decay rates of the modes. The dependence of the GP resonance frequency on the graphene Fermi level enables electrical tuning of the hybridized system.

The hybridization between the guided-mode resonance and the graphene plasmonic mode produces an avoided-crossing response that enables large electrically tunable phase excursions while maintaining nearly uniform reflectance. The spectral behavior of the hybridized resonances and the resulting phase modulation are analyzed in detail in the following subsection.

A. Hybridized Resonances and Phase Modulation

The spectral response of the coupled system is shown in Fig. 2. Figure 2(a) shows the magnitude of the reflection coefficient $|r_p|$ as a function of the wavelength and the graphene Fermi level E_f. As E_f increases, the resonance frequency of the graphene plasmonic mode shifts according to $\omega_{\text{GP}} \propto \sqrt{E_f}$, allowing the GP mode to sweep across the almost fixed GMR resonance frequency. When the two resonances approach each other spectrally, a clear avoided crossing appears in the reflection map, indicating strong hybridization between the two modes.

The corresponding phase response is shown in Fig. 2(b). Near the anticrossing region, the hybridized modes undergo a rapid phase transition as the graphene Fermi level is varied. Importantly, this phase transition occurs while the reflection amplitude remains relatively uniform, which is a

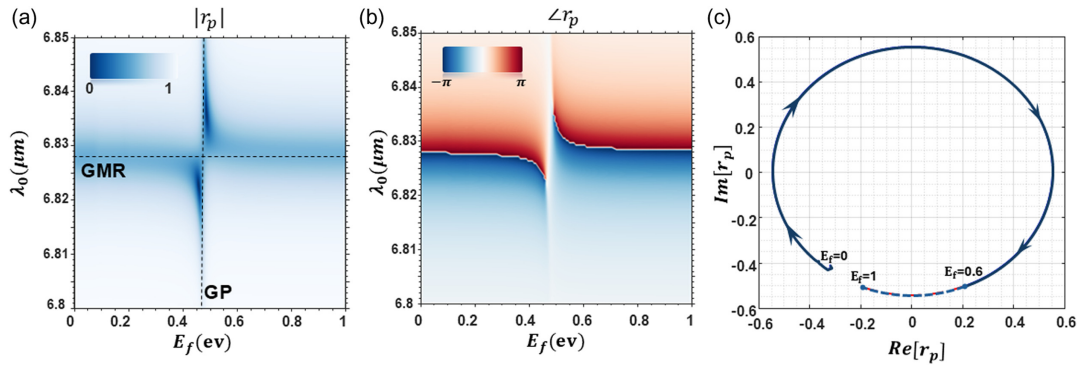


Fig. 2. Hybridized resonance behavior and phase modulation. (a) Magnitude of the reflection coefficient $|r_p|$ as a function of the wavelength λ and the graphene Fermi level E_f. The avoided crossing between the guided-mode resonance (GMR) and the graphene plasmonic (GP) mode indicates strong hybridization. (b) Corresponding phase of the reflection coefficient $\angle r_p$. Near the anticrossing region, the hybridized modes exhibit a rapid phase transition as E_f varies. (c) Trajectory of the complex reflection coefficient r_p in the complex plane at the operating wavelength $\lambda_0 = 6.83$ μm as the graphene Fermi level is tuned. The nearly circular trajectory demonstrates continuous $\sim 2\pi$ phase modulation while maintaining nearly uniform reflection amplitude.

direct consequence of the overcoupled nature of the dielectric resonance.

A more intuitive representation of this behavior is obtained by examining the trajectory of the complex reflection coefficient. Figure 2(c) shows the evolution of r_p in the complex plane as the graphene Fermi level is tuned. The resulting trajectory traces an approximately circular path, indicating that the reflection magnitude remains nearly constant while the phase continuously wraps around the circle. This behavior confirms that the metasurface supports nearly complete 2π phase modulation with minimal variation in the reflection amplitude.

The ability to achieve large phase excursions while maintaining relatively uniform reflection amplitude originates from the hybridization of the overcoupled guided-mode resonance with the electrically tunable graphene plasmonic mode. The electrostatic feasibility of this tuning is discussed in Note S2 of Supplement 1. Because the radiative decay rate of the GMR exceeds its dissipative loss rate, the hybridized modes remain predominantly radiatively coupled near the anticrossing region. As a result, the phase of the reflection coefficient varies strongly while the amplitude remains nearly unchanged. Material losses also play an important role in the efficiency and phase stability of the proposed graphene-based mid-infrared metasurface. In the RCWA simulations, absorption losses are included through the complex optical constants of silicon, gold, and HfO_2, while graphene loss is included through the complex Kubo surface conductivity. Graphene plasmon damping, determined by the carrier relaxation time, broadens the graphene plasmonic resonance and can reduce the strength of its coupling with the guided-mode resonance. Increased damping can therefore weaken the avoided crossing, reduce the minimum reflection amplitude, distort the complex reflection trajectory, and degrade the uniformity of the 2π phase modulation. Dielectric and metallic absorption contribute to the dissipative decay rate of the resonant system and reduce the overall reflection and sideband-conversion efficiencies. Nevertheless, the present design is intentionally engineered in the overcoupled regime, where the radiative decay rate of the guided-mode resonance exceeds the dissipative loss rate. This condition improves the robustness of the phase response to moderate material losses because the hybridized resonance can redistribute the phase while maintaining a relatively uniform reflection amplitude. This property forms the physical foundation for the wavefront engineering functionalities demonstrated in the following sections.

B. Wavefront Engineering and Beam Steering

Having established continuous phase modulation through the hybridized GMR–GP resonance, we next demonstrate static wavefront engineering by designing a metasurface supercell that steers the reflected beam into a specified diffraction order. The beam-steering functionality is implemented using a supercell composed of three distinct unit cells, each operating at a different graphene Fermi level to realize the required phase profile at the operating wavelength $\lambda_0 = 6.83$ μm.

For a metasurface with a linear phase gradient under normal incidence, the steering angle follows the generalized grating relation:

$$\sin\theta_m = \frac{m\lambda_0}{\Lambda_{sc}}, \tag{2}$$

where m denotes the diffraction order and Λ_{sc} is the supercell period. In our design, the supercell consists of three unit cells, yielding $\Lambda_{sc} = 3\Lambda = 3 \times 2596$ nm $= 7.788$ μm. For the first diffraction order ($m = +1$), Eq. (2) predicts a steering angle of

$$\theta_{+1} = \sin^{-1}\left(\frac{\lambda_0}{\Lambda_{sc}}\right) = \sin^{-1}\left(\frac{6.83}{7.788}\right) \approx 61.3^\circ. \tag{3}$$

In the forward-design approach, the desired phase profile is constructed from a linear phase ramp across the supercell. The target phase at the nth unit cell is given by

$$\phi_n^{\text{target}} = \phi_{\text{ref}} + \frac{2\pi m}{\Lambda_{sc}} x_n \pmod{2\pi}, \tag{4}$$

where x_n denotes the position of the nth unit cell and ϕ_{ref} is an arbitrary reference phase. Since the three unit cells are separated by the lattice period Λ, this expression reduces to

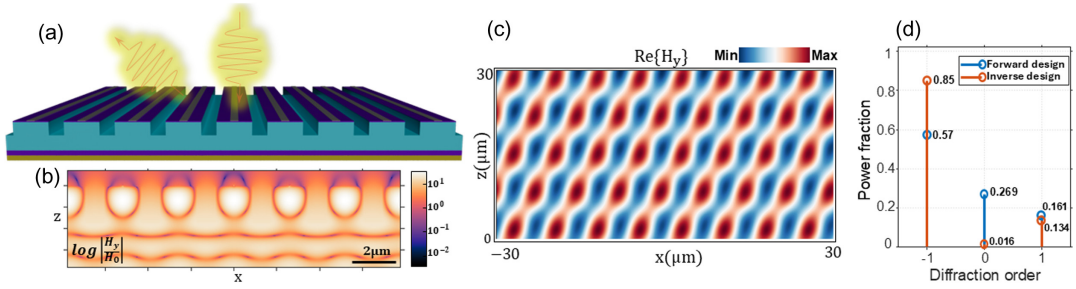


Fig. 3. Wavefront engineering and beam steering using electrically programmable unit cells. (a) Schematic illustration of the metasurface implementing a linear phase gradient across the array. The required phase values are obtained by electrically tuning the graphene Fermi level according to the unit-cell phase response shown in Fig. 2. (b) Simulated near-field distribution $\log(|H_y/H_0|)$ of the metasurface. Three unit cells form one supercell implementing the discrete phase steps $\{0, 2\pi/3, 4\pi/3\}$, and two supercells are shown in the simulation domain. (c) Far-field reflected field distribution $\mathrm{Re}\{H_y\}$ demonstrating beam steering into the first diffraction order at an angle of approximately 61°, consistent with the generalized grating relation. (d) Power distribution among diffraction orders for the forward-designed and inverse-designed metasurfaces. Inverse-design optimization compensates for inter-cell coupling and significantly enhances the first-order diffraction efficiency.

$$\phi_n^{\text{target}} = \phi_{\text{ref}} + \frac{2\pi m}{3}(n-1) \pmod{2\pi}, \qquad n = 1, 2, 3. \tag{5}$$

For $m = +1$, the ideal phase values are therefore separated by $2\pi/3$, corresponding to the discrete phase sequence $\{0, 2\pi/3, 4\pi/3\}$ up to a global phase offset. These target phases are implemented by selecting the graphene Fermi levels that best reproduce the required phase values from the unit-cell phase response shown in Fig. 2. Using this forward-design procedure, the three unit cells were chosen with $E_f = [0.1, 0.5145, 0.555]$ eV.

While the forward-designed supercell successfully produces the desired phase gradient, it does not explicitly account for the residual electromagnetic coupling between neighboring unit cells. Because the hybridized resonances retain a finite spatial coherence length, the optical response of each cell is slightly perturbed by the presence of adjacent cells. As a result, the forward-designed structure achieves a first-order diffraction efficiency of approximately 57%.

To mitigate this effect, we further optimized the graphene Fermi levels directly at the supercell level using an inverse-design procedure. The optimized values $E_f = [0.1, 0.517, 0.546]$ eV compensate for the inter-cell coupling and significantly improve the diffraction performance.

Figure 3 shows the resulting beam-steering behavior of the optimized metasurface. Figure 3(b) shows the simulated near-field magnitude distribution $|H_y/H_0|$ obtained using an in-house developed rigorous coupled-wave analysis (RCWA) solver, where H_0 denotes the incident magnetic-field amplitude. The field map clearly reveals the formation of the engineered phase front across the metasurface supercell.

The corresponding far-field response is shown in Fig. 3(c) through the reflected field $\mathrm{Re}\{H_y\}$. The far-field distribution confirms that the reflected beam is predominantly redirected into the first diffraction order at approximately 61°, in excellent agreement with the prediction of Eq. (2).

To quantify performance, the diffraction efficiency into order m is defined as

$$\eta_m = \frac{P_m}{\sum_q P_q}, \tag{6}$$

where P_m denotes the reflected power carried by the mth diffraction order and the denominator represents the total reflected power summed over all propagating diffraction orders. The power distribution among diffraction orders for both the forward-designed and inverse-designed supercells is shown in Fig. 3(d). While the forward design achieves a first-order diffraction efficiency of 57%, the inverse-designed metasurface increases this efficiency to 85% by suppressing parasitic diffraction channels and concentrating a larger fraction of the reflected power into the desired +1 diffraction order.

These results confirm that the hybridized GMR–graphene platform not only enables nearly complete 2π phase control at the unit-cell level but also supports efficient wavefront engineering when residual inter-cell interactions are properly taken into account.

C. Spatiotemporal Modulation and Single-Sideband Beam Steering

In addition to conventional active metasurfaces, where various tuning mechanisms are exploited in a stationary fashion to modify performance metrics, spatiotemporal metasurfaces leverage optical properties continuously varying in time, enabling functionalities that exceed the limits of static optical systems. In this context, time can be incorporated as an additional degree of freedom in the problem space to enable a wide range of applications across classical and quantum platforms, including directional light scattering and amplification in single dynamic particles [32–34], manipulation of quantum light through control of the time-varying density of electromagnetic states [35,36], nonreciprocal light–matter interactions in spatiotemporal metasurfaces [37,38], and amplification and zero-threshold lasing enabled by the opening of a momentum bandgap in photonic time crystals [39,40].

Of substantial importance is the frequency mixing attribute of spatiotemporal metasurfaces with temporally periodic optical characteristics, whereby the interaction of a monochromatic incident wave with the metasurface gives rise to multiple frequency components corresponding to integer harmonics of the modulation frequency [41,42]. In such systems, meticulous control of the temporal modulation profile not only enables

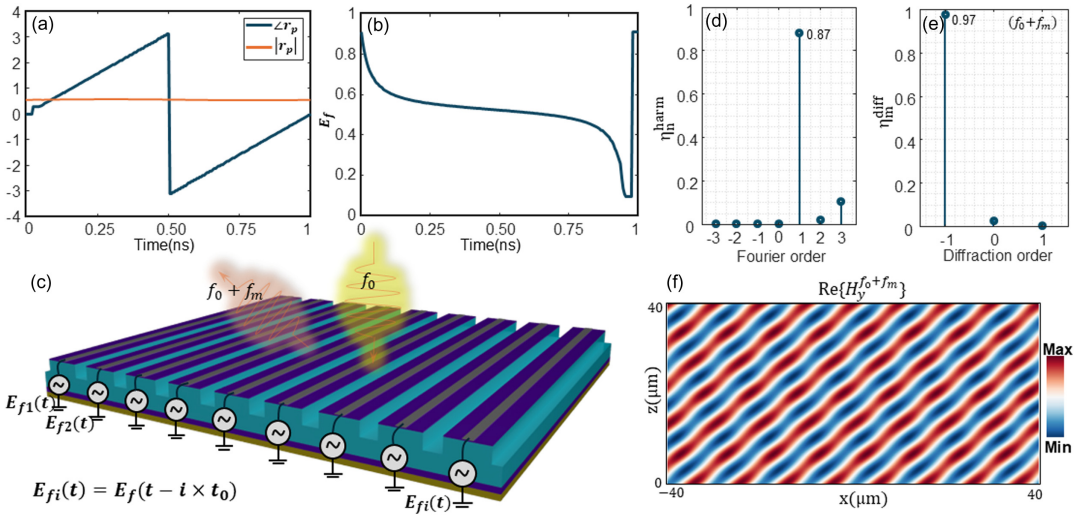


Fig. 4. Spatiotemporal modulation and single-sideband beam steering. (a) Time evolution of the reflection phase $\angle r_p$ (blue) and the amplitude $|r_p|$ (orange), showing sawtooth temporal phase modulation with nearly constant amplitude. (b) Corresponding time-dependent graphene Fermi level $E_f(t)$ required to generate the target phase modulation. (c) Schematic of the spatiotemporal modulation scheme, where progressive temporal phase delays are applied between adjacent pixels to create the required spatiotemporal phase gradient. (d) Fourier spectrum of the generated temporal harmonics, showing dominant conversion to the first-order sideband. (e) Diffraction-efficiency distribution among spatial diffraction orders at the up-converted frequency. (f) Far-field distribution of the up-converted sideband $\mathrm{Re}\{H_y^{f_0+f_m}\}$, demonstrating directional beam steering.

the selective generation of desired harmonics but also allows for dispersionless control of the spatial wavefront. More specifically, as opposed to conventional active metasurfaces, where wavefront shaping is prominently achieved by tuning resonant responses that are inherently dispersive, the phase response in spatiotemporally modulated metasurfaces can be engineered through temporal modulation, enabling phase control that is largely independent of the intrinsic material or structural dispersion [43,44]. To achieve such dispersionless wavefront shaping, continuous 2π phase coverage across the metasurface is a prerequisite.

The continuous phase tunability of the hybridized GMR–GP resonance enables dynamic spatiotemporal modulation of the metasurface. By periodically modulating the graphene Fermi level in time, the reflection coefficient becomes a time-varying quantity, which produces frequency sidebands around the carrier frequency f_0. Under appropriate modulation conditions, the metasurface can be engineered to generate a single sideband, thereby enabling frequency conversion and wavefront control simultaneously.

Following the single-sideband suppressed-carrier (SSB-SC) condition described in Ref. [16], the temporal reflection coefficient must satisfy

$$|r(t)| = |r_{+1}|, \qquad \angle r(t) = \omega_m t + \alpha, \tag{7}$$

where $\omega_m = 2\pi f_m$ is the modulation angular frequency and α is an arbitrary phase offset. The linear phase evolution is periodically wrapped into the 2π interval, producing a sawtooth temporal phase profile. Under this condition, the temporal modulation generates a single up-modulated sideband at frequency $f_0 + f_m$.

To realize this behavior, the desired temporal phase profile is defined as

$$\phi(t) = (\omega_m t + \alpha) \bmod 2\pi. \tag{8}$$

Using the quasi-static phase response $\Phi(E_f) = \angle r_p(E_f)$ obtained from the unit-cell simulations in Fig. 2, the required time-dependent graphene Fermi level is obtained through the inverse mapping:

$$E_f(t) = \Phi^{-1}[\phi(t)]. \tag{9}$$

The resulting temporal evolution of the reflection coefficient is shown in Fig. 4(a), where the amplitude $|r_p|$ (orange curve) remains nearly constant while the phase $\angle r_p$ (blue curve) follows the desired sawtooth waveform. The corresponding time-dependent graphene Fermi level is shown in Fig. 4(b). In this work, the modulation frequency is chosen as $f_m = 1$ GHz.

The quasi-static mapping between the instantaneous Fermi level and the reflection response is valid under the adiabatic condition, which requires the modulation frequency to be much smaller than the optical carrier frequency. For the operating wavelength $\lambda_0 = 6.83$ μm, the carrier frequency is

$$f_0 = \frac{c}{\lambda_0} \approx 43.9 \text{ THz}, \tag{10}$$

which satisfies $f_m \ll f_0$. Therefore, the instantaneous reflection response can be approximated by the stationary metasurface response corresponding to the instantaneous graphene Fermi level.

Spatiotemporal beam steering is achieved by introducing a progressive temporal phase delay between adjacent unit cells. For the ith cell, the modulation signal is written as

$$E_{F,i}(t) = E_f(t - it_0), \tag{11}$$

where t_0 represents the delay between neighboring elements. The corresponding modulation phase shift is

$$\Delta\alpha = \omega_m t_0. \tag{12}$$

For a desired steering angle θ_b, the required phase delay between adjacent pixels is given by

$$\Delta\alpha = k_0 \Lambda_{\mathrm{uc}} \sin\theta_b, \tag{13}$$

where $k_0 = 2\pi/\lambda_0$ and Λ_{uc} is the unit-cell period. The resulting spatiotemporal modulation scheme is shown in Fig. 4(c), where progressive time delays are applied across the metasurface to realize the required spatiotemporal phase gradient.

The spectral distribution of the generated temporal harmonics is shown in Fig. 4(d). The modulation produces a dominant first-order sideband at frequency $f_0 + f_m$. The harmonic conversion efficiency is defined as

$$\eta_{\mathrm{harm}} = \frac{P(f_0 + f_m)}{\sum_n P(f_0 + n f_m)}, \tag{14}$$

where $P(f_0 + n f_m)$ denotes the power contained in the nth temporal harmonic. The resulting conversion efficiency reaches approximately 87%, indicating efficient single-sideband generation.

In addition to frequency conversion, the spatiotemporal phase gradient enables directional radiation of the generated sideband. The diffraction efficiency at the sideband frequency is shown in Fig. 4(e) and defined as

$$\eta_{\mathrm{diff}} = \frac{P_{m=+1}(f_0 + f_m)}{\sum_q P_q(f_0 + f_m)}, \tag{15}$$

where $P_m(f_0 + f_m)$ denotes the power of diffraction order m at the up-converted frequency. The first diffraction order carries approximately 97% of the total sideband power, indicating highly efficient beam formation.

Finally, the far-field distribution of the generated sideband is shown in Fig. 4(f), where the reflected field $\mathrm{Re}\{H_y^{f_0+f_m}\}$ forms a well-defined beam propagating at approximately 41°. This result confirms that the proposed spatiotemporally modulated metasurface can simultaneously achieve efficient single-sideband frequency conversion and dynamic beam steering (see Note S6 of Supplement 1 for details).

3. METHODS

Full-wave electromagnetic simulations were performed using an in-house rigorous coupled-wave analysis (RCWA) solver under TM-polarized normal incidence. The metasurface was modeled as a one-dimensional periodic grating. The structure is periodic along the x direction with period Λ, invariant along the y direction, and stratified along the z direction. The silicon bars and graphene strips are assumed to extend uniformly along y. The excitation is a normally incident TM-polarized plane wave, with the magnetic field oriented along y. Periodic/Floquet boundary conditions are applied along x, translational invariance is assumed along y, and outgoing-wave radiation conditions are imposed along z. For beam-steering calculations, a supercell was defined along the x direction by grouping multiple unit cells within one RCWA period, and the reflected power was decomposed into the propagating spatial Fourier diffraction orders. The complex reflection coefficient $r_p(\lambda, E_f)$ was calculated as a function of the free-space wavelength λ and the graphene Fermi level E_f. To ensure numerical convergence in the presence of the deeply subwavelength graphene strip, 801 spatial Fourier harmonics ($2 \times 400 + 1$) were retained in RCWA calculations.

Monolayer graphene was modeled using the Kubo formalism, including both intraband and interband conductivity contributions. In the RCWA implementation, graphene was treated as a thin conductive layer with an effective permittivity derived from its surface conductivity. A nominal graphene thickness of 1 nm was used in the numerical model while preserving the correct sheet conductivity. The graphene carrier mobility was taken as $\mu_s = 3500\ \mathrm{cm^2\,V^{-1}\,s^{-1}}$. The dispersive optical constants of silicon, gold, and HfO_2 were taken from experimentally measured mid-infrared datasets.

The metasurface geometry was optimized using a genetic algorithm at the operating wavelength $\lambda_0 = 6.83$ μm. The figure of merit was defined to maximize the achievable phase excursion across the graphene Fermi-level tuning range while penalizing variations in the reflection amplitude. The optimized parameter set included the period Λ, silicon bar width w_{bar}, bar thickness t_{bar}, core thickness t_{core}, spacer thickness t_{ins}, and graphene width w_g, while the HfO_2 thickness was fixed at 30 nm.

For the spatiotemporal modulation analysis, the adiabatic approximation was used because the modulation frequency $f_m = 1$ GHz is much smaller than the optical carrier frequency. The time-dependent reflection response was sampled using quasi-static RCWA at instantaneous values of the graphene Fermi level $E_f(t)$, and the sideband amplitudes were extracted through Fourier decomposition of the resulting temporal reflection coefficient. Diffraction efficiencies were calculated from the reflected power carried by the corresponding spatial Fourier orders. Temporal coupled-mode theory (TCMT) was used as an interpretive framework to explain the hybridized resonance behavior, whereas all reported optical responses and device-performance metrics were obtained from RCWA simulations. We emphasize that TCMT is used only as an analytical and interpretive model to explain the avoided-crossing behavior and the role of radiative and dissipative decay rates; all quantitative reflection spectra, phase responses, diffraction efficiencies, and sideband conversion efficiencies are obtained from the full-wave RCWA simulations.

4. CONCLUSION

In summary, we have demonstrated an electrically tunable mid-infrared metasurface that achieves nearly complete 2π phase modulation through strong coupling between an overcoupled guided-mode resonance (GMR) and a graphene plasmonic (GP) mode. Electrical tuning of the graphene Fermi level shifts the plasmonic resonance across the fixed GMR frequency, producing strong mode hybridization that redistributes the phase while maintaining a minimum reflection amplitude of 0.57 at the operating wavelength $\lambda_0 = 6.83$ μm. An inverse-design optimization framework compensates for residual inter-cell coupling, improving beam-steering efficiency from 57% to 85%. Leveraging the large electrically tunable phase range, we further demonstrate spatiotemporal modulation enabling

Table 1. Comparison with Representative Active and Spatiotemporally Modulated Metasurfaces

Work	Platform	Modulation Scheme	Range	Phase	Efficiency/Key Metric
Sherrott *et al.* [21]	Graphene–gold	Electrical gating	Mid-IR	>230°	N/R for conversion efficiency
Shirmanesh *et al.* [20]	Dual-gated ITO	Dual-gate tuning	1550 nm	>300°	N/R for conversion efficiency
Forouzmand and Mosallaei [18]	ITO–GMR	Electrical tuning	Near-IR	Large tuning	Tunable amplitude/phase response
Kim *et al.* [22]	qBIC–graphene	Avoided crossing	Mid-IR	Full 2π	Nearly uniform reflection, ∼0.65
Sabri *et al.* [16]	Spatiotemporal metasurface	Time modulation	Near-IR	2π ramp	Normalized SSB conversion >99%
Salary *et al.* [43]	Time-modulated graphene	Temporal modulation	THz	2π harmonic	Harmonic wavefront control
This work	**GMR–graphene**	**Single-gate avoided crossing + time modulation**	**Mid-IR**	$\sim 2\pi$	85% **beam steering;** ∼87% **harmonic conversion;** and ∼97% **sideband diffraction**

single-sideband suppressed-carrier beam steering with a harmonic conversion efficiency of $\eta_{SSB} \approx 87\%$ and a first-order diffraction efficiency of ∼97%. Table 1 compares representative active and spatiotemporally modulated metasurfaces in terms of modulation scheme, operating wavelength, phase range, and reported efficiency metrics.

The current implementation operates under TM polarization, and extending the hybridization mechanism to TE polarization or dual-polarization operation remains an open direction for future work. Achieving polarization-independent 2π phase modulation would further broaden the applicability of the platform for arbitrary vectorial wavefront control.

Looking forward, the demonstrated combination of full phase tunability, near-uniform reflectance, and spatiotemporal programmability positions this platform as a promising building block for several compelling mid-infrared applications. Integration into focal plane arrays would enable large-scale programmable apertures for dynamic beam steering and real-time holographic displays. The electrically addressable phase response at the pixel level is also well suited for active computational metasurface architectures capable of performing optical analog processing and on-chip optical computing directly in the mid-infrared. Furthermore, the access to the molecular fingerprint region offered by the $\lambda_0 = 6.83$ μm operating wavelength suggests potential utility in label-free molecular fingerprinting and spectroscopic sensing, where dynamic wavefront control could enhance both spatial resolution and spectral selectivity. Collectively, these directions motivate further experimental realization and scaling of the proposed platform toward practical mid-infrared photonic systems.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

Supplemental document. See Supplement 1 for supporting content.

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